



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

MATHEMATICS SYLLABUS A
PAPER 2

4075/2
2 hours 30 minutes

SPECIMEN PAPER

Additional materials:
Mathematical Instruments
Mathematical Tables/Non programmable Electronic Calculator
Answer booklet

INSTRUCTIONS TO CANDIDATES

Write your Name, Centre number and Candidate number in the spaces provided on the answer booklet.

Answer **all questions** in Section A and **any four questions** from Section B.

If you use more than one booklet, fasten them together.

All working must be clearly shown on the same sheet as the rest of the answer.

Omission of essential working will result in loss of marks.

If the degree of accuracy is not specified in the question and if the answer is not exact, the answer should be given correct to three significant figures. Answers in degrees should be given correct to one decimal place.

Mathematical tables and/or Non-programmable electronic calculators may be used to evaluate explicit numerical expressions.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

This specimen paper consists of 9 printed pages.

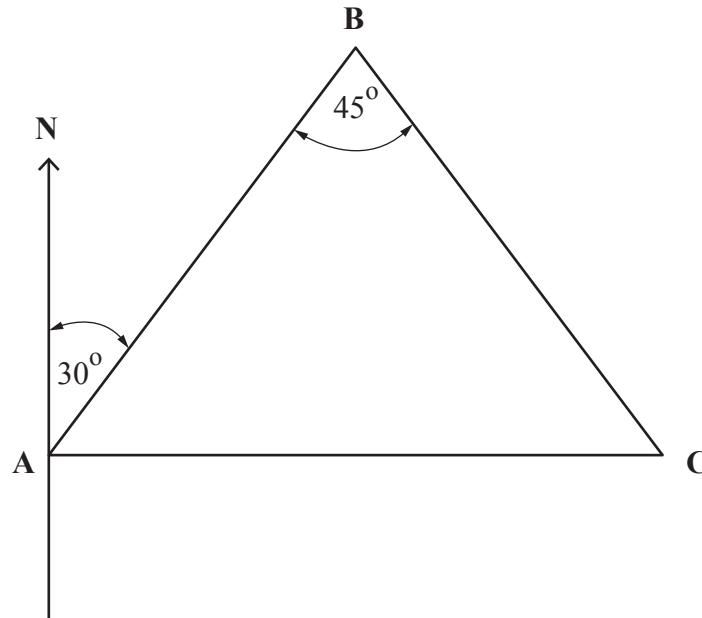
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Section A [52 Marks]

Answer **all** questions in this section

- 1** **(a)** Evaluate each of the following when
 $a = 1$, $b = -2$ and $c = 3$
- (i)** $a + b + c$ [2]
- (ii)** $a^2 - b$. [2]
- (b)** Express $\frac{5}{x-2} + \frac{1}{3}$ as a single fraction in its simplest form. [3]
- (c)** Solve the following equation $x^2 - 7x + 10 = 0$. [3]
- 2** **(a)** Find the Highest Common Factor (H.C.F) of :
- $3^3 \times 5^2 \times 11^2$
- $3 \times 5^4 \times 11$
- $3^2 \times 5^2 \times 11^3$
- Leave the answer in index form. [2]
- (b)** **(i)** Express 108; 216 and 648 as products of their prime factors. [3]
- (ii)** Find the Lowest Common Multiple of 108; 216 and 648. [2]
- (c)** **(i)** Find the Highest Common Factor of $24x^2y^3$ and $36x^3y^2$. [1]
- (ii)** Find the Lowest Common Multiple (L.C.M) of
 $24x^2y^3$ and $36x^3y^2$. [2]

- 3 The diagram below shows three points A, B and C on level ground. B is on a bearing of N 30° E from point A and $\angle ABC = 45^\circ$.



- (a) State the three-figure bearing of **B** from **A**. [1]
- (b) Find the bearing of:
- (i) **A** from **B**, [1]
- (ii) **B** from **C**. [1]
- (c) Two similar cuboids have corresponding sides in the ratio 4:6.
- (i) Express the ratio in its simplest form. [1]
- (ii) Find the ratio of the volumes of the cuboids. [2]
- (iii) If the volume of the larger cuboid is 81 cm^3 ,
find the volume of the smaller cuboid. [2]
- (iv) Find the mass of the larger cuboid if its density is 0.5 g / cm^3 . [2]

- 4 (a) Evaluate

$$-5 + (-6) \div (+3) \quad [2]$$

- (b) A rectangle has a length of 10 cm and a width of 6 cm rounded off to the nearest centimetre.

Calculate the least possible

- (i) perimeter of the rectangle, [3]
- (ii) area of the rectangle. [2]
- (c) The cost of 3 pens is 45 c.
- Calculate the cost of 50 pens, giving the answer in dollars. [3]

5 **ANSWER THE WHOLE OF THIS QUESTION ON A SHEET OF PLAIN PAPER**

Use ruler and compasses only for all constructions and show clearly all construction lines and arcs.

All constructions should be done on a single diagram.

A tuckshop, T is located 7,5 km due west of the school, S. The tuckshop is also 5 km due south of the church, C.

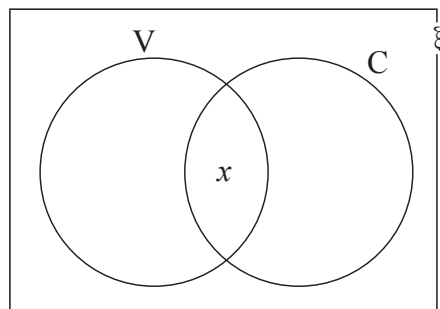
- (a) Using a scale of 1 cm: 1 km, construct a diagram to show the positions of T, S and C. [5]
- (b) Find the distance from the school to the church, in km. [2]
- (c) Bisect (i) line TS, [2]
- (ii) $\widehat{CTS}$. [2]
- (d) Mark and label P, the intersection of the two bisectors. [1]

Section B [48 Marks]

Answer any **four** questions from this section.

Each question carries **12** marks

- 6 (a) The Universal set, ξ is such that $\xi = \{1; 2; 3; \dots; 10\}$
- List all the
- (i) even numbers, [2]
- (ii) prime numbers. [2]
- (b) Z is a set of integers.
- N is a set of natural numbers.
- C is a set of prime numbers between 10 and 20.
- (i) Identify a finite set, from the sets above. [1]
- (ii) Express sets Z and N in set notation. [1]
- (c) There are 25 learners in a class. 16 play volleyball, 10 play chess, x learners play both games. 5 players do not play any of the given games.
- Let ξ be set of learners in the class,
- V be the set of learners who play volleyball
- C be the set of learners who play chess.
- (i) Copy and complete the Venn diagram below, by showing the number of elements in each set.



[3]

- (ii) Find the value of x . [3]

7 The points P and Q have coordinates at $(5; -6)$ and $(-3; -8)$ respectively.

- (a) Calculate the gradient of line PQ. [2]

- (b) Find the equation of the line PQ. [2]

ANSWER THE WHOLE OF THIS QUESTION ON A SHEET OF GRAPH PAPER.

Use a scale of 2 cm to represent 1 unit on both axes, for $-2 \leq x \leq 2$ and $-2 \leq y \leq 1$.

- (c) Draw, on the same axes, the graphs of

(i) $x + 2y = 1$

(ii) $3x - 2y = 3$

[4]

- (d) Use the graph to solve the simultaneous equations:

$$x + 2y = 1$$

$$3x - 2y = 3$$

[2]

- (e) Write down the y -intercept and the x -intercept of the line $3x - 2y = 3$ [2]

8 Given that $\mathbf{A} = \begin{pmatrix} 3 & 2 \\ 5 & 4 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} n & 2 \\ 4 & -2 \end{pmatrix}$, $\mathbf{C} = \begin{pmatrix} x & 4 \\ 5 & 3 \end{pmatrix}$ and

$$\mathbf{D} = \begin{pmatrix} k+1 & 6 \\ 2 & k+2 \end{pmatrix}, \text{ find the}$$

- (a) inverse of matrix $\mathbf{A}$, [3]

- (b) value of n if matrix $\mathbf{B}$ is singular, [3]

- (c) value of x when determinant of $\mathbf{C}$ is -5 , [3]

- (d) two values of k if matrix $\mathbf{D}$ is singular. [3]

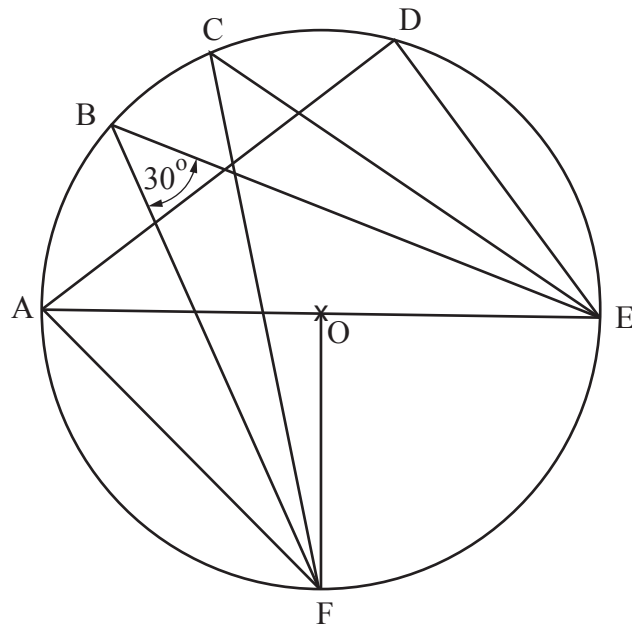
- 9 The marks obtained by 36 learners in a Mathematics test that was marked out of 5 were as follows:

1	3	2	3	4	2	1	3	0
5	3	0	1	4	0	4	4	3
3	4	3	1	3	4	3	1	2
1	3	4	0	4	3	2	5	3

- (a) Show the data on a
- (i) tally table, [1]
- (ii) frequency table. [2]
- (b) Calculate the
- (i) mean, [2]
- (ii) mode, [1]
- (iii) median. [2]
- (c) **Answer the whole of this question on a sheet of graph paper.**
- Use a scale of **2 cm** to represent **1** unit on both axes.
- Draw a bar graph for the information. [3]
- (d) If a learner is selected at random, find the probability that the learner selected scored a mark less than 3. [1]

- 10 (a) Express $\frac{35}{x-4} + \frac{5}{x^2-9x+20}$ as a single fraction in its simplest form. [4]
- (b) (i) Express 3 969 as a product of its prime factors. [1]
- (ii) Hence or otherwise find $\sqrt{3\,969}$. [1]
- (c) Factorise completely:
- (i) $(x - y)(4x + 2y) - (x - y)^2$ [2]
- (ii) $x^2 + 5x - 6$ [2]
- (d) Solve the equation $\frac{x-2}{5} = 2\frac{1}{2}$. [2]

11



In the diagram A, B, C, D, E and F are points on the circumference of a circle centre O. $\widehat{EBF} = 30^\circ$

- (a) (i) Name an angle equal to $\widehat{ECF}$, [2]
- (ii) Name an angle equal to $\widehat{AFC}$. [1]
- (b) State the size of
- (i) $\widehat{ADE}$, [1]
- (ii) $\widehat{ECF}$, [1]
- (iii) $\widehat{EAF}$, [1]
- (c) Calculate $\widehat{EOF}$. [2]
- (d) (i) Solve the inequality $5x + 7 \leq 2x - 5$. [3]
- (ii) Show the solution in (d)(i) on a number line. [1]

12 A stove has a marked price of \$720.00.

- (a)** The stove can be bought on cash basis. If bought on cash basis a discount of 20% is offered.

Calculate the

(i) discount offered, [2]

(ii) discounted price. [2]

- (b)** The stove can be bought on lay-by for six months.

Calculate the monthly lay-by instalment. [2]

- (c)** The stove can also be bought on hire purchase. A deposit of 30% of the marked price is paid in cash. Then 12 equal monthly instalments of \$80.00 each are paid.

Calculate the

(i) deposit paid, [2]

(ii) total cost of the stove by hire-purchase. [4]